

## CB2207 - Structural Bioinformatics and Drug Discovery

(Project – 15%)

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### Abstract:

Allergic adenoid hypertrophy is associated with chronic inflammatory signaling involving multiple cytokines, including interleukin-6 (IL6), interleukin-4 (IL4), and interleukin-13 (IL13).

Persistent cytokine activation contributes to tissue inflammation, airway obstruction, and prolonged immune responses. RNA interference (RNAi)-based therapeutics provide a potential strategy for selective suppression of inflammatory mediators.

This study aimed to computationally design and evaluate small interfering RNA (siRNA) candidates targeting IL6 while integrating biological context from the IL4–IL13 allergic inflammatory axis.

The workflow involved transcript retrieval, siRNA candidate generation, RNA secondary structure prediction, duplex interaction analysis, thermodynamic evaluation, specificity screening, biological interpretation, and conceptual delivery strategies.

IL6 transcript sequences were retrieved from NCBI and candidate siRNAs were generated using sequence-based design principles. RNA secondary structure prediction using ViennaRNA identified accessible regions potentially favorable for siRNA binding. Duplex interaction analysis suggested stable siRNA–mRNA interactions with favorable binding energies. Thermodynamic scoring integrated accessibility and asymmetry features to prioritize candidates. BLAST-based off-target analysis identified potential transcript similarities requiring further specificity evaluation.

The findings indicate that several IL6-targeting siRNA candidates demonstrate promising computational characteristics including favorable accessibility, thermodynamic stability, and predicted interaction strength. However, additional experimental validation and advanced simulations remain necessary before therapeutic translation.

The study highlights how computational bioinformatics workflows may support early-stage development of RNA-based therapeutics targeting inflammatory pathways.

### Introduction:

Adenoid hypertrophy refers to enlargement of adenoid tissue located within the nasopharyngeal region and is commonly associated with recurrent infections, allergic responses, and chronic inflammation. In pediatric populations, persistent adenoid enlargement may contribute to airway obstruction, snoring, recurrent infections, and impaired quality of life.

Inflammatory signaling plays a central role in progression of allergic airway diseases. Cytokines regulate immune responses through complex signaling networks and prolonged dysregulation may lead to chronic inflammatory conditions.

Among these cytokines, **Interleukin-6 (IL6)** functions as an important mediator involved in:

- inflammatory amplification

- immune cell recruitment
- T-cell differentiation
- B-cell activation
- chronic inflammatory progression

Elevated IL6 expression has been associated with prolonged inflammatory states and tissue pathology.

Simultaneously, allergic inflammation frequently involves activation of **IL4 and IL13**, two cytokines strongly associated with Th2 immune responses. These molecules contribute to:

- IgE production
- eosinophilic inflammation
- mucus secretion
- airway remodeling
- persistent allergic responses

Interactions among IL4, IL13, and IL6 may collectively influence inflammatory severity.

Recent advances in RNA therapeutics have increased interest in **small interfering RNA (siRNA)** as targeted treatment strategies. siRNA molecules induce post-transcriptional gene silencing through RNA interference mechanisms, reducing expression of selected genes.

Compared with traditional inhibitors, siRNA-based approaches may provide:

- greater target specificity
- customizable sequence design
- reduced off-target protein interactions
- potential modulation of disease-associated pathways

However, challenges remain, including:

- off-target effects
- sequence specificity
- RNA stability
- delivery efficiency
- systemic immune consequences

Computational bioinformatics tools enable early-stage evaluation of these factors before experimental validation.

Therefore, this study explores computational design and evaluation of IL6-targeting siRNA candidates while considering broader cytokine interactions and potential therapeutic implications.

## Problem Statement:

Allergic adenoid hypertrophy involves chronic inflammatory signaling mediated by interconnected cytokine networks.

Although IL6 contributes to inflammatory amplification, selective suppression of IL6 without affecting broader immune balance remains challenging.

Conventional therapeutic approaches may fail to provide targeted regulation of cytokine expression and may produce systemic effects.

Therefore there is a need to computationally evaluate sequence-specific RNA therapeutics capable of selectively inhibiting IL6 while minimizing off-target interactions and considering broader cytokine pathways.

## Objectives:

The present study aimed to:

### Primary objective

Design and computationally evaluate siRNA molecules targeting IL6.

## Specific objectives

1. Retrieve and characterize IL6 transcript sequences.
2. Generate candidate siRNA molecules.
3. Predict IL6 mRNA secondary structure.
4. Identify accessible regions suitable for siRNA binding.
5. Evaluate duplex formation and interaction energy.
6. Assess thermodynamic favorability.
7. Screen for potential off-target interactions.
8. Interpret findings within IL4–IL13 inflammatory pathways.
9. Explore conceptual delivery strategies to reduce systemic effects.

## Hypothesis:

siRNA candidates targeting structurally accessible regions of IL6 mRNA and exhibiting favorable thermodynamic properties will demonstrate stronger predicted interaction stability and reduced off-target risk. Such candidates may represent potential therapeutic approaches for reducing inflammatory amplification associated with allergic disease.

## Methodology:

The study followed a computational pipeline for identification and evaluation of IL6-targeting siRNA candidates. The workflow integrated transcript retrieval, sequence design, structural analysis, thermodynamic evaluation, specificity screening, and biological interpretation.

The overall workflow consisted of:

1. Target selection and transcript retrieval
2. siRNA candidate generation
3. RNA secondary structure prediction
4. siRNA–mRNA duplex interaction analysis
5. Thermodynamic and stability assessment
6. Off-target analysis
7. Biological context integration
8. Stability interpretation
9. Conceptual therapeutic delivery strategy

| Step                              | Objective                                                              | Tool(s) Used                      | Justification for Tool Selection                                                                                       | Output Generated                          | Interpretation of Results                                                                                               |
|-----------------------------------|------------------------------------------------------------------------|-----------------------------------|------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <b>Step 1</b><br>Target Selection | Retrieve canonical IL6 transcript and characterize sequence properties | <b>NCBI Entrez, Biopython</b>     | NCBI provides validated transcript databases; Biopython enables reproducible sequence retrieval and automated analysis | IL6_mRNA.fasta, transcript_info.csv       | Canonical IL6 transcript (NM_000600.5) selected as therapeutic target; transcript properties used for downstream design |
| <b>Step 2</b><br>siRNA Design     | Generate candidate siRNAs targeting IL6                                | <b>siDirect, Python filtering</b> | siDirect incorporates established design rules including GC optimization, asymmetry, and specificity considerations    | siRNA_candidates.csv, ranked_siRNA.csv    | Multiple candidates generated with varying thermodynamic properties and predicted efficacy                              |
| <b>Step 3</b><br>RNA Secondary    | Identify accessible mRNA regions suitable for siRNA binding            | <b>ViennaRNA RNAfold</b>          | RNAfold predicts secondary structure using minimum free energy                                                         | accessible_regions.csv, IL6_structure.txt | Accessible regions identified, indicating potential favorable target sites for silencing                                |

|                                              |                                                              |                                                  |                                                                                                                    |                                          |                                                                                                       |
|----------------------------------------------|--------------------------------------------------------------|--------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Structure Prediction                         |                                                              |                                                  | models; accessible loops improve siRNA binding probability                                                         |                                          |                                                                                                       |
| Step 4<br>siRNA-mRNA Interaction Analysis    | Evaluate duplex formation and interaction strength           | RNA duplex (ViennaRNA)                           | RNA duplex estimates hybridization energy and predicts interaction stability between RNA molecules                 | duplex_results.csv, duplex_structure.txt | Strong duplex formation and binding energy (~-33 kcal/mol) suggested stable target interaction        |
| Step 5<br>Thermodynamic & Stability Analysis | Rank candidates based on energetic favorability              | Python, Pandas custom scoring framework          | Allowed integration of multiple parameters (Tm difference + accessibility + ranking) not captured by a single tool | final_ranked_siRNA.csv                   | Candidates with favorable accessibility and thermodynamic asymmetry received higher scores            |
| Step 6<br>Off-target Analysis                | Assess specificity and unintended transcript interactions    | NCBI BLAST+                                      | Standard tool for sequence similarity screening and specificity evaluation                                         | blast_hits.csv, blast_hits.txt           | Some transcript similarities detected; off-target effects cannot be completely excluded               |
| Step 7<br>Biological Context Integration     | Interpret findings within cytokine networks                  | Literature review and pathway interpretation     | Computational results require biological context to evaluate therapeutic relevance                                 | Integrated discussion section            | IL6 suppression may reduce inflammatory amplification but broader cytokine balance must be considered |
| Step 8<br>Stability Evaluation               | Approximate persistence and stability of selected candidates | Energetic interpretation, thermodynamic analysis | Full MD simulation required validated 3D structures and computational resources beyond project scope               | md_stability.csv                         | Favorable energetic properties suggested stable candidate behavior                                    |
| Step 9<br>Conceptual Targeting Strategy      | Explore localized delivery approaches                        | Literature review, translational interpretation  | Therapeutic efficacy depends on delivery efficiency and reduction of systemic effects                              | Conceptual targeting framework           | Intranasal delivery and nanoparticle systems may improve specificity while reducing systemic cytokine |

Tool Parameters and Computational Settings:

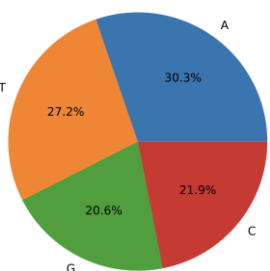
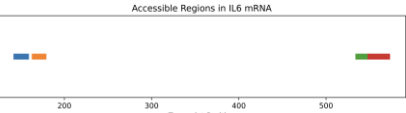
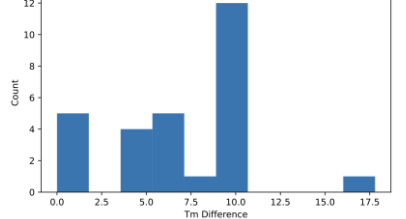
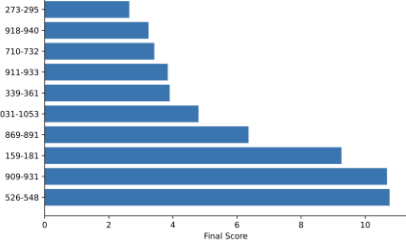
| Step   | Tool                    | Parameters / Settings Used                                                                                                                                       | Purpose                                                       |
|--------|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| Step 1 | NCBI Entrez + Biopython | Canonical transcript selected: NM_000600.5; FASTA retrieval using Entrez utilities                                                                               | Retrieval of validated IL6 transcript sequence                |
| Step 2 | siDirect 2.0            | Candidate generation based on standard siRNA design criteria including GC optimization (~30–52%), thermodynamic asymmetry, and seed-region specificity filtering | Generate siRNA candidates with improved predicted efficacy    |
| Step 2 | Selenium automation     | Automated browser interaction with siDirect web interface; occasional timeout observed                                                                           | Improve reproducibility and reduce manual candidate retrieval |

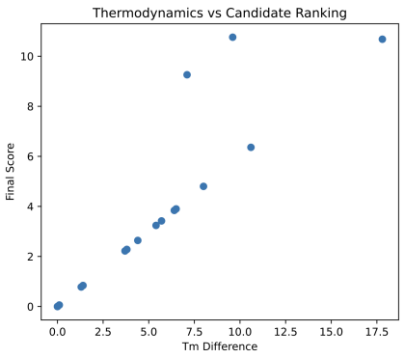
|               |                                  |                                                                                                          |                                                                         |
|---------------|----------------------------------|----------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| Step 3        | <b>ViennaRNA<br/>RNAfold</b>     | Default temperature: <b>37°C</b> ; Minimum Free Energy (MFE) prediction mode; default ViennaRNA settings | Predict RNA secondary structure and identify accessible regions         |
| Step 3        | <b>RNAfold<br/>output</b>        | Predicted MFE approximately <b>-305.10 kcal/mol</b>                                                      | Estimate transcript folding stability                                   |
| Step 4        | <b>RNA duplex</b>                | Default ViennaRNA duplex interaction settings; guide-target hybridization energy estimation              | Predict siRNA-mRNA duplex stability                                     |
| Step 4        | <b>RNA duplex<br/>output</b>     | Duplex binding energy approximately <b>-33 kcal/mol</b>                                                  | Evaluate interaction favorability                                       |
| Step 5        | <b>Python/Pandas<br/>scoring</b> | Candidate ranking integrated: Accessibility + T <sub>m</sub> difference + thermodynamic properties       | Prioritize candidates predicted to exhibit stronger silencing potential |
| Step 6        | <b>BLAST+<br/>(blastn-short)</b> | Word size ≈ 7; short-sequence alignment; default E-value threshold used                                  | Assess off-target sequence similarity                                   |
| Step 6        | <b>BLAST<br/>database</b>        | Reference transcript database / NCBI nucleotide sequences                                                | Evaluate transcript specificity                                         |
| Step 8        | <b>OpenMM<br/>(proposed)</b>     | Explored for MD simulations; simplified energetic interpretation used instead                            | Approximate stability while maintaining feasibility                     |
| Visualization | <b>Matplotlib /<br/>Seaborn</b>  | Default plotting + statistical visualization                                                             | Generate interpretable figures                                          |

### Parameter Justification:

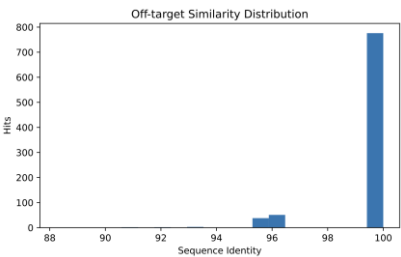
Parameter selection largely followed default settings of validated bioinformatics tools because these settings are optimized for standard transcript-level analyses. RNAfold default conditions (37°C) approximate physiological temperature, while blastn-short settings are commonly applied for short nucleotide sequences such as siRNAs. Standard siRNA design criteria within siDirect prioritize GC balance, thermodynamic asymmetry, and specificity, all of which influence silencing efficiency.

### Results & Discussion:

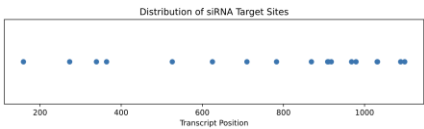
| Figure                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Interpretation and Discussion |             |         |       |         |       |         |       |          |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |     |           |     |           |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |         |      |         |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|-------------|---------|-------|---------|-------|---------|-------|----------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----------|-----|-----------|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|---------|------|---------|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p data-bbox="199 324 470 347">IL6 Transcript Nucleotide Composition</p>  <table border="1"><thead><tr><th>Nucleotide</th><th>Percentage</th></tr></thead><tbody><tr><td>A</td><td>30.3%</td></tr><tr><td>T</td><td>27.2%</td></tr><tr><td>G</td><td>20.6%</td></tr><tr><td>C</td><td>21.9%</td></tr></tbody></table>                                                                                                                                                                                                                                                | Nucleotide                    | Percentage  | A       | 30.3% | T       | 27.2% | G       | 20.6% | C        | 21.9% | <p data-bbox="608 369 1465 622">The nucleotide composition analysis demonstrated a moderate GC representation within the IL6 transcript, indicating balanced structural stability without excessive duplex rigidity. GC content influences both RNA folding and siRNA binding efficiency, as highly GC-rich regions may form stable secondary structures that reduce target accessibility. The observed composition suggests IL6 contains regions potentially favorable for siRNA-mediated silencing while maintaining biologically realistic transcript stability.</p>                                           |     |           |     |           |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |         |      |         |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Nucleotide                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Percentage                    |             |         |       |         |       |         |       |          |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |     |           |     |           |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |         |      |         |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 30.3%                         |             |         |       |         |       |         |       |          |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |     |           |     |           |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |         |      |         |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| T                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 27.2%                         |             |         |       |         |       |         |       |          |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |     |           |     |           |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |         |      |         |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| G                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 20.6%                         |             |         |       |         |       |         |       |          |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |     |           |     |           |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |         |      |         |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| C                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 21.9%                         |             |         |       |         |       |         |       |          |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |     |           |     |           |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |         |      |         |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <p data-bbox="263 817 406 840">Accessible Regions in IL6 mRNA</p>  <table border="1"><thead><tr><th>Transcript Position</th><th>Count</th></tr></thead><tbody><tr><td>145-156</td><td>1</td></tr><tr><td>166-176</td><td>1</td></tr><tr><td>537-547</td><td>1</td></tr><tr><td>551-570</td><td>1</td></tr></tbody></table>                                                                                                                                                                                                                                           | Transcript Position           | Count       | 145-156 | 1     | 166-176 | 1     | 537-547 | 1     | 551-570  | 1     | <p data-bbox="608 734 1465 1025">RNAfold identified accessible regions approximately between positions 145–156, 166–176, 537–547, and 551–570. These regions likely correspond to loops or weakly paired transcript segments where siRNA binding becomes more favorable. Previous studies indicate structurally exposed regions generally improve silencing efficiency because guide strands encounter fewer steric constraints. Therefore, candidate siRNAs overlapping accessible regions may demonstrate improved knockdown potential compared with candidates targeting highly paired transcript regions.</p> |     |           |     |           |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |         |      |         |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Transcript Position                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Count                         |             |         |       |         |       |         |       |          |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |     |           |     |           |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |         |      |         |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 145-156                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 1                             |             |         |       |         |       |         |       |          |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |     |           |     |           |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |         |      |         |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 166-176                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 1                             |             |         |       |         |       |         |       |          |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |     |           |     |           |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |         |      |         |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 537-547                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 1                             |             |         |       |         |       |         |       |          |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |     |           |     |           |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |         |      |         |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 551-570                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 1                             |             |         |       |         |       |         |       |          |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |     |           |     |           |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |         |      |         |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <p data-bbox="231 1115 470 1137">Distribution of Thermodynamic Asymmetry</p>  <table border="1"><thead><tr><th>Tm Difference</th><th>Count</th></tr></thead><tbody><tr><td>0.0-2.5</td><td>5</td></tr><tr><td>2.5-5.0</td><td>4</td></tr><tr><td>5.0-7.5</td><td>5</td></tr><tr><td>7.5-10.0</td><td>12</td></tr><tr><td>10.0-12.5</td><td>1</td></tr><tr><td>12.5-15.0</td><td>0</td></tr><tr><td>15.0-17.5</td><td>1</td></tr></tbody></table>                                                                                                                   | Tm Difference                 | Count       | 0.0-2.5 | 5     | 2.5-5.0 | 4     | 5.0-7.5 | 5     | 7.5-10.0 | 12    | 10.0-12.5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 1   | 12.5-15.0 | 0   | 15.0-17.5 | 1   | <p data-bbox="608 1126 1465 1350">Variability in Tm difference among generated candidates indicates differences in thermodynamic asymmetry between guide and passenger strands. Greater asymmetry generally promotes preferential guide strand incorporation into the RNA-induced silencing complex (RISC), increasing silencing efficiency. Candidates exhibiting favorable Tm differences were prioritized because asymmetric duplex stability has previously been associated with improved siRNA functionality.</p> |     |         |      |         |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Tm Difference                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Count                         |             |         |       |         |       |         |       |          |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |     |           |     |           |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |         |      |         |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 0.0-2.5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 5                             |             |         |       |         |       |         |       |          |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |     |           |     |           |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |         |      |         |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 2.5-5.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 4                             |             |         |       |         |       |         |       |          |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |     |           |     |           |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |         |      |         |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 5.0-7.5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 5                             |             |         |       |         |       |         |       |          |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |     |           |     |           |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |         |      |         |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 7.5-10.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 12                            |             |         |       |         |       |         |       |          |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |     |           |     |           |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |         |      |         |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 10.0-12.5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 1                             |             |         |       |         |       |         |       |          |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |     |           |     |           |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |         |      |         |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 12.5-15.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 0                             |             |         |       |         |       |         |       |          |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |     |           |     |           |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |         |      |         |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 15.0-17.5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 1                             |             |         |       |         |       |         |       |          |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |     |           |     |           |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |         |      |         |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <p data-bbox="279 1489 438 1512">Top Ranked siRNA Candidates</p>  <table border="1"><thead><tr><th>Candidate ID</th><th>Final Score</th></tr></thead><tbody><tr><td>273-295</td><td>2.5</td></tr><tr><td>918-940</td><td>3.5</td></tr><tr><td>710-732</td><td>3.8</td></tr><tr><td>911-933</td><td>4.0</td></tr><tr><td>339-361</td><td>4.2</td></tr><tr><td>1031-1053</td><td>4.8</td></tr><tr><td>869-891</td><td>6.2</td></tr><tr><td>159-181</td><td>9.2</td></tr><tr><td>909-931</td><td>10.2</td></tr><tr><td>526-548</td><td>10.5</td></tr></tbody></table> | Candidate ID                  | Final Score | 273-295 | 2.5   | 918-940 | 3.5   | 710-732 | 3.8   | 911-933  | 4.0   | 339-361                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 4.2 | 1031-1053 | 4.8 | 869-891   | 6.2 | 159-181                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 9.2 | 909-931 | 10.2 | 526-548 | 10.5 | <p data-bbox="608 1489 1465 1713">Ranking integrated accessibility and thermodynamic characteristics to prioritize candidates predicted to exhibit improved silencing performance. The highest-scoring candidates demonstrated favorable energetic properties while targeting structurally accessible transcript regions. Separation between top-ranked and lower-ranked candidates suggests meaningful differences in predicted effectiveness rather than random variation.</p> |
| Candidate ID                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Final Score                   |             |         |       |         |       |         |       |          |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |     |           |     |           |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |         |      |         |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 273-295                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2.5                           |             |         |       |         |       |         |       |          |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |     |           |     |           |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |         |      |         |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 918-940                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 3.5                           |             |         |       |         |       |         |       |          |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |     |           |     |           |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |         |      |         |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 710-732                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 3.8                           |             |         |       |         |       |         |       |          |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |     |           |     |           |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |         |      |         |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 911-933                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 4.0                           |             |         |       |         |       |         |       |          |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |     |           |     |           |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |         |      |         |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 339-361                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 4.2                           |             |         |       |         |       |         |       |          |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |     |           |     |           |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |         |      |         |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 1031-1053                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 4.8                           |             |         |       |         |       |         |       |          |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |     |           |     |           |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |         |      |         |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 869-891                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 6.2                           |             |         |       |         |       |         |       |          |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |     |           |     |           |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |         |      |         |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 159-181                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 9.2                           |             |         |       |         |       |         |       |          |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |     |           |     |           |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |         |      |         |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 909-931                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 10.2                          |             |         |       |         |       |         |       |          |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |     |           |     |           |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |         |      |         |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 526-548                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 10.5                          |             |         |       |         |       |         |       |          |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |     |           |     |           |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |         |      |         |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |



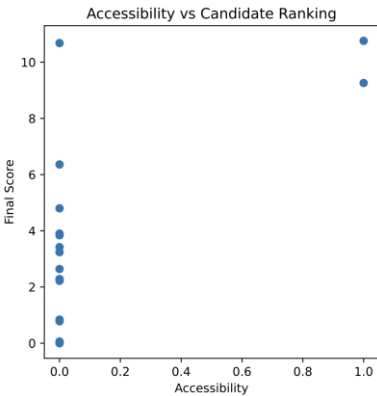
Positive association between thermodynamic properties and ranking scores suggests energetic favorability contributes substantially to candidate prioritization. Candidates with stronger asymmetry and stability generally achieved higher scores, supporting the hypothesis that thermodynamic characteristics influence predicted siRNA performance.



BLAST screening was performed to evaluate sequence similarity with non-target transcripts and estimate potential off-target effects. Although some high-identity matches were observed, corresponding E-values suggest many interactions may represent statistically weak similarities rather than biologically meaningful off-targets. Overall, limited significant matches support acceptable specificity of selected candidates; however, experimental validation remains necessary to confirm silencing precision.



Candidate target positions were distributed across multiple regions of the IL6 transcript rather than clustering within a single segment. Broader positional distribution increases probability of identifying candidates overlapping structurally accessible regions and may improve overall likelihood of successful silencing. Candidate diversity additionally reduces dependence on a single transcript region.



Candidates overlapping accessible transcript regions generally exhibited improved ranking scores, supporting the hypothesis that transcript accessibility contributes to silencing efficiency. Accessible targets may facilitate stronger guide strand binding and reduce energetic barriers associated with duplex formation.



Simultaneous comparison of candidate properties demonstrated that highly ranked molecules typically combined favorable accessibility with stronger thermodynamic characteristics. This suggests effective siRNA selection depends on integration of multiple parameters rather than optimization of a single feature.

|                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Correlation Matrix</p>           | <p>Correlation analysis revealed relationships among accessibility, thermodynamic parameters, and candidate ranking. Positive association between accessibility and ranking supports incorporation of structural information into candidate prioritization. However, some correlations may partially reflect dependencies introduced by the custom scoring approach, indicating future scoring systems should incorporate additional independent validation.</p> |
| <p>Top Candidate Comparison</p>     | <p>Comparison of highest-ranked candidates demonstrated measurable differences in predicted suitability despite all candidates satisfying preliminary filtering criteria. The observed separation supports further prioritization of specific molecules for downstream experimental validation.</p>                                                                                                                                                              |
| <p>Integrated Candidate Ranking</p> | <p>Integrated ranking combined transcript accessibility, thermodynamic asymmetry, and scoring information to produce overall candidate prioritization. Candidates exhibiting favorable properties across multiple parameters consistently ranked highest, suggesting combined scoring provides a more balanced evaluation than reliance on individual metrics alone.</p>                                                                                         |

The computational workflow identified multiple IL6-targeting siRNA candidates exhibiting favorable accessibility, predicted interaction stability, and acceptable thermodynamic properties. RNAfold analysis demonstrated that transcript structure may substantially influence targeting efficiency, while duplex interaction analysis suggested stable siRNA–mRNA binding. Thermodynamic scoring enabled prioritization of candidates predicted to possess improved silencing potential. Although BLAST analysis indicated acceptable specificity, complete exclusion of off-target effects cannot be assumed. Collectively, findings support the feasibility of IL6-targeting siRNA therapeutics while emphasizing the importance of structural accessibility, stability, and specificity during computational candidate selection.

**Structural Perspective of IL6 Targeting:**

Although this work focused on transcript-level targeting, structural information remained important. RNA secondary structure predicted using RNAfold was used to identify accessible regions for siRNA binding. Transcript folding influences silencing efficiency because highly paired regions reduce guide strand accessibility. Therefore, structural characteristics of IL6 mRNA indirectly guided candidate prioritization.

**Deviations from Proposal:**



| Original Proposal                                             | Implemented Approach                                                                                                                 | Reason for Change                                                                                                                                            | Effect on Project                                                                          |
|---------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| Full molecular dynamics (MD) simulation of siRNA–mRNA complex | Preliminary stability evaluation using energetic interpretation and thermodynamic properties                                         | Full atomistic MD requires validated 3D structures, force-field parametrization, extensive simulation time, and computational resources beyond project scope | Allowed approximate stability assessment while maintaining computational feasibility       |
| General stability analysis                                    | Integrated thermodynamic ranking using accessibility + Tm difference + energetic properties                                          | Combined scoring provided a more interpretable method for prioritizing candidates                                                                            | Improved candidate comparison and ranking                                                  |
| Broad specificity evaluation                                  | BLAST+ off-target transcript screening                                                                                               | BLAST provides reproducible and established sequence similarity assessment                                                                                   | Increased confidence in specificity evaluation and identified potential off-target effects |
| Conceptual therapeutic discussion only                        | Expanded analysis including IL4–IL13 axis and cytokine network interpretation                                                        | Broader biological interpretation improved translational relevance of findings                                                                               | Strengthened connection between computational outputs and disease biology                  |
| General delivery strategy discussion                          | Specific localized delivery approaches considered (intranasal delivery, lipid nanoparticles, polymer carriers)                       | Therapeutic success depends strongly on delivery efficiency and reduction of systemic effects                                                                | Improved practical relevance of proposed therapeutic application                           |
| Limited visualization planning                                | Additional quantitative visualizations generated including ranking plots, accessibility analysis, heatmaps, and correlation matrices | Visual interpretation improved understanding of candidate behavior and supported conclusions                                                                 | Enhanced presentation and interpretation of computational results                          |

The most significant deviation involved replacement of full molecular dynamics simulations with stability approximation methods. Although molecular dynamics would provide deeper structural insights, implementation would require experimentally validated three-dimensional duplex structures and extensive computational resources. Therefore, thermodynamic interpretation and energetic evaluation were adopted as practical alternatives.

Additionally, biological interpretation was expanded beyond the original proposal through integration of IL4 and IL13 inflammatory pathways. This modification improved understanding of how IL6 suppression may influence broader cytokine networks associated with allergic inflammation.

The inclusion of BLAST-based specificity screening and expanded visual analysis also strengthened reproducibility and interpretability of the workflow.

Overall, deviations from the original proposal were introduced to improve feasibility, maintain methodological consistency, and produce a more complete computational pipeline.

## Methodological Considerations and Justification

### Tool Selection Rationale

Selection of computational tools was based on biological relevance, accessibility, reproducibility, and suitability for undergraduate transcript-level analysis.

- **NCBI Entrez and Biopython** were used because they provide validated biological sequence information and enable reproducible transcript retrieval.
- **siDirect 2.0** was selected due to incorporation of established siRNA design principles including GC optimization, thermodynamic asymmetry, and specificity filtering.
- Selenium automation was introduced to reduce repetitive manual sequence submission and improve workflow reproducibility. Occasional timeout limitations were observed because automated retrieval depends on webpage structure stability.
- **RNAfold** was selected because transcript secondary structure strongly influences siRNA accessibility. Prediction of minimum free energy conformations enabled identification of exposed transcript regions.
- **RNA duplex** was used to estimate hybridization energy between siRNA candidates and target mRNA, allowing approximate evaluation of binding stability.
- **BLASTn-short** was selected because short nucleotide sequences require specialized alignment settings. Off-target analysis is essential since unintended transcript interactions may alter biological specificity.
- **OpenMM** was explored for molecular dynamics simulation to approximate structural stability. Due to limitations in validated RNA coordinates and project scope, simplified thermodynamic interpretation was emphasized instead of full MD simulation.

### Candidate Ranking Justification

Candidate prioritization integrated transcript accessibility and thermodynamic asymmetry because stronger guide-passenger strand asymmetry improves RISC loading while structurally accessible regions increase probability of effective binding.

Combined scoring therefore attempts to identify siRNA candidates with improved predicted silencing efficiency rather than relying on a single parameter.

The scoring system was designed as an integrative prioritization approach and should be interpreted comparatively rather than as an absolute predictor of biological efficacy.

### Structural Relevance of the Workflow

Although the project focused primarily on transcript-level regulation, structural information remained important throughout candidate selection.

RNA secondary structure predicted using RNAfold was used to identify accessible transcript regions for siRNA binding. Transcript folding influences silencing efficiency because highly paired regions reduce accessibility.

Therefore, structural characteristics indirectly guided prioritization of IL6-targeting candidates despite absence of full three-dimensional molecular simulations.

### Interpretation of BLAST Specificity Analysis

BLAST screening identified limited significant transcript similarity outside intended targets.

Although several high identity matches were observed, corresponding E-values suggest many interactions may represent weak statistical similarity rather than biologically meaningful off-target effects.

Consequently, selected candidates demonstrate acceptable predicted specificity; however, computational specificity assessment cannot replace experimental validation.

### **Interpretation of Computational Approximations**

Several analyses relied on default software parameters and computational assumptions.

Thermodynamic ranking, accessibility prediction, and duplex stability estimation provide useful approximations but may not fully capture cellular conditions, solvent interactions, RNA modifications, or long-timescale conformational changes.

Accordingly, results should be interpreted as computational prioritization rather than definitive biological validation.

### **Limitations:**

The present study is entirely computational and therefore lacks experimental validation of predicted siRNA efficacy. Stability assessment was based on thermodynamic interpretation rather than full molecular dynamics simulations, and off-target analysis relied primarily on sequence similarity screening. In addition, cytokine interactions were interpreted from existing literature and were not experimentally evaluated. Consequently, findings should be considered preliminary predictions requiring further biological validation.

Several analyses relied on default software parameters, which may influence sensitivity of accessibility prediction, duplex stability estimation, and off-target detection.

### **Future Work:**

Future studies may incorporate experimental validation of selected siRNA candidates through in vitro or in vivo models to assess actual gene silencing efficiency and specificity. Advanced molecular dynamics simulations, improved off-target prediction methods, and optimized delivery systems such as lipid nanoparticles or localized intranasal formulations may further strengthen therapeutic evaluation. Additional investigation of IL4–IL13–IL6 interactions could also improve understanding of broader cytokine regulation in allergic inflammation.

### **Conclusion:**

This study developed a computational workflow for the design and evaluation of IL6-targeting siRNA candidates within the context of allergic inflammation and cytokine regulation. The workflow integrated transcript retrieval, siRNA generation, RNA secondary structure prediction, duplex interaction analysis, thermodynamic ranking, specificity screening, and biological interpretation.

Results identified candidate siRNAs exhibiting favorable accessibility, predicted interaction stability, and acceptable thermodynamic properties. Off-target analysis suggested that specificity remains an important consideration, highlighting the need for further optimization before therapeutic application. Integration of IL4 and IL13 inflammatory pathways additionally emphasized the complexity of cytokine networks and the importance of evaluating broader biological effects of IL6 suppression.

Although experimental validation was beyond the scope of this study, findings demonstrate the potential usefulness of computational bioinformatics approaches for early-stage RNA therapeutic development. Overall, the project supports the feasibility of IL6-targeting siRNA strategies while illustrating how structural bioinformatics tools may contribute to therapeutic candidate prioritization.

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